

Internship Project — Week 3 and Week 4

Project Title: End to End Sales Forecasting and Demand Intelligence System

Submitted by: Indrayani Prashant Mude

St. Vincent Pallotti College of Engineering and Technology, Nagpur

Problem Statement: Every retail and e-commerce company — from Walmart and Amazon to D-Mart and Flipkart — lives and dies by one question: "How much of each product will we sell next month, and will we have enough stock to meet that demand?" Getting this wrong in either direction costs crores — overstock wastes storage and capital, understock loses sales and customers.

This Machine Learning system predicts future product demand, detects unusual sales spikes or drops, segments products by demand pattern, and presents everything through a deployed interactive dashboard.

Executive Summary

This report analyzes four years of retail sales data to help improve inventory planning and financial decision-making. Historical sales patterns were studied, unusual sales behavior was identified, products were grouped based on demand characteristics, and future sales were forecasted using multiple machine learning and statistical models. Among the models evaluated, XGBoost produced the most accurate forecasts (MAPE: 14.1%), making it the recommended model for future planning. The analysis indicates continued business growth, highlights a few unusual sales periods that require attention, and provides stocking recommendations for different product groups. Implementing these insights can help reduce stock shortages, avoid excess inventory, and improve demand planning over the next quarter.

1. Key Findings from Exploratory Data Analysis

Sales Trends

- Overall sales increased steadily throughout the four-year period.

- Strong yearly seasonality was observed, with recurring peaks during similar months each year.
- Sales growth accelerated during later years, indicating improving business performance.

Model Performance

Three forecasting models were evaluated.

Model	MAE	RMSE	MAPE
SARIMA	20275.307039	21062.311698	21.5%
Prophet	40970.325067	53868.948681	40.0%
XGBoost	15009.498125	20566.478787	14.1%

Business Interpretation

- XGBoost generated the lowest prediction error and most reliable forecasts.
- SARIMA successfully modeled seasonality but was less accurate.
- Prophet captured long-term trends but struggled with sudden fluctuations.

Therefore, XGBoost is recommended for operational forecasting and inventory planning.

2. Three-Month Sales Forecast with Confidence Ranges

The XGBoost forecasting model, which achieved the highest accuracy (MAPE: **14.1%**), predicts that overall sales will be approximately **87,393** in January 2019, **50,846** in February 2019, and **30,142** in March 2019.

Although the XGBoost model provides the final business forecasts, the SARIMA model was used to estimate forecast uncertainty. Based on historical sales patterns, actual sales are expected to fluctuate within a reasonable range around the predicted values. The uncertainty widens for forecasts further into the future, meaning that while the predicted sales represent the most likely outcomes, external factors such as promotions, holidays, supplier delays, and changing customer demand may cause actual sales to differ.

Category-wise forecasting indicates that **Furniture** is expected to experience the strongest growth (22.6K → 32.7K → 35.8K), **Office Supplies** will maintain stable demand with gradual growth (25.4K → 25.7K → 27.7K), while **Technology** is expected to grow initially before declining in the third month (24.1K → 26.6K → 22.4K). Regionally, the **West** is projected to show the highest increase in sales, whereas the **East** is expected to maintain steady growth.

3. Top 3 Anomalies Detected and Their Likely Causes

The anomaly detection module, using **Isolation Forest** and **Rolling Z-Score**, identified several unusual sales observations.

1. Sudden High Sales Spike

Likely Cause: Seasonal promotions, festive discounts, or bulk customer purchases that generated unusually high sales compared to normal business patterns.

2. Significant Drop in Sales

Likely Cause: Temporary inventory shortages, reduced customer demand, or operational disruptions leading to lower-than-expected sales.

3. Short-Term Sales Fluctuations

Likely Cause: Regional demand variations, one-time marketing campaigns, or data irregularities that caused temporary deviations from the normal sales trend.

These anomalies should be reviewed individually before making inventory or procurement decisions, as they may not represent regular business behavior.

4. Product Demand Segmentation Findings and Recommended Stocking Strategy

Products were segmented into demand groups using **K-Means clustering** based on sales performance and demand characteristics.

Segment	Findings	Recommended Stocking Strategy
High-Demand Products	Consistently high sales and strong customer demand.	Maintain higher safety stock, prioritize replenishment, and closely monitor inventory levels to avoid stockouts.
Medium-Demand Products	Stable and predictable demand throughout the year.	Maintain moderate inventory levels with regular replenishment based on forecasted demand.
Low-Demand Products	Low or irregular sales with higher inventory holding risk.	Adopt a lean stocking approach by ordering smaller quantities and replenishing only when required to reduce storage costs.

This demand-based segmentation enables better inventory optimization and more efficient allocation of warehouse space and procurement budgets.

5. Business Recommendations

1. Use XGBoost as the Primary Forecasting Model

Data Backing

- Lowest forecasting error among all evaluated models.
- **MAPE: 14.1%**
- Lower MAE and RMSE than SARIMA and Prophet.

Business Benefit

More accurate demand forecasting will improve procurement planning, reduce inventory costs, and minimize stock shortages.

2. Increase Inventory Allocation for the Furniture Category and West Region

Data Backing

- Furniture sales are forecast to increase from **22.6K** to **35.8K** over the next three months.
- West region sales are projected to increase from **17.7K** to **29.8K**.

Business Benefit

Increasing stock availability in these high-growth segments will help meet rising customer demand and prevent lost sales due to stockouts.

3. Use Anomaly Detection Before Procurement Planning

Data Backing

The anomaly detection models identified unusual sales spikes and drops that could distort forecasting if treated as normal demand.

Business Benefit

Reviewing anomalies before procurement decisions helps prevent unnecessary overstocking after temporary demand spikes or understocking following short-term sales declines.

6. Risk / Limitation of the System

The forecasting system primarily relies on historical sales data and does not currently account for external business factors such as competitor pricing, promotional campaigns, economic conditions, holidays, weather, or unexpected supply chain disruptions. Consequently, while the forecasts provide reliable planning estimates, significant external events may cause actual sales to differ from the predicted values. Future versions of the system can improve forecasting accuracy by incorporating these external variables into the predictive models.